

Experiment: Student Utility Program

Menu-Driven Java Program — Basics, Conditionals, Loops & Methods

1. Question

Develop a Java-based “Student Utility Program” using a menu-driven approach that performs the following operations:

Part A: Java Basics

- Accept student name, roll number, and marks.
- Calculate total and percentage.
- Display the student result.

Part B: Conditional Statements

- Check whether a number is even or odd.
- Find the largest of three numbers.
- Display grade based on percentage using if-else.
- Display the day of the week using switch.

Part C: Looping Statements

- Print a multiplication table.
- Display numbers from 1 to N.
- Calculate the sum of the first N natural numbers.
- Generate the Fibonacci series.

Part D: Methods

Create and call parameterized, return-type methods to:

- Find factorial.
- Check whether a number is prime.
- Find the maximum of two numbers.
- Calculate the area of a circle.

2. Steps

- 1 Start the program.
- 2 Import the Scanner class.
- 3 Create a static Scanner object for user input.
- 4 Create a menu containing options from 1 to 14.
- 5 Read the user's choice.
- 6 Use a do-while loop to repeatedly display the menu until the user selects Exit.
- 7 Use a switch statement to perform the selected operation.

Part A

- Accept student name, roll number, and three subject marks.
- Calculate total marks.
- Calculate percentage.
- Display the student result.

Part B

- Accept a number and check whether it is even or odd.

- Accept three numbers and find the largest.
- Accept percentage and display the appropriate grade.
- Accept a number from 1 to 7 and display the corresponding day.

Part C

- Accept a number and limit and print its multiplication table.
- Accept N and display numbers from 1 to N.
- Calculate the sum of the first N natural numbers using a while loop.
- Generate Fibonacci terms using a do-while loop.

Part D

- Create the parameterized factorial() method and return the factorial.
- Create the parameterized isPrime() method and return true or false.
- Create the parameterized maximum() method and return the larger number.
- Create the parameterized circleArea() method and return the area.
- Call these methods from main().
- Display the results.
- Select option 14 to exit the program.
- Close the Scanner.
- Stop the program.

3. Algorithm

Main Algorithm

- 1 Start.
- 2 Initialize the Scanner.
- 3 Display the Student Utility Program menu.
- 4 Read the user's choice.
- 5 Execute the selected operation using switch.
- 6 Repeat the menu using do-while until choice 14.
- 7 Close the Scanner.
- 8 Stop.

Part A – Student Result

- 1 Read student name.
- 2 Read roll number.
- 3 Read three subject marks.
- 4 Calculate total = mark1 + mark2 + mark3.
- 5 Calculate percentage = total / 3.
- 6 Display student details and result.

Part B – Conditional Statements

Even/Odd:

- 1 Read a number.
- 2 If number % 2 == 0, display Even.
- 3 Otherwise, display Odd.

Largest of Three:

- 1 Read three numbers.
- 2 Compare all three numbers.
- 3 Store the largest number.
- 4 Display it.

Grade:

- 1 Read percentage.
- 2 Check percentage using if-else.
- 3 Display A+, A, B, C, D, E, or F.

Day:

- 1 Read a number from 1 to 7.
- 2 Use switch.
- 3 Display the corresponding day.

Part C – Loops

Multiplication Table:

- 1 Read number and limit.
- 2 Repeat from 1 to limit.
- 3 Calculate number $\times$ i.
- 4 Display each result.

Numbers 1 to N:

- 1 Read N.
- 2 Check whether N is positive.
- 3 Use a for loop from 1 to N.

- 4 Display each number.

Sum of Natural Numbers:

- 1 Read N.
- 2 Initialize sum = 0 and i = 1.
- 3 Add i to sum.
- 4 Increment i.
- 5 Repeat until i > N.
- 6 Display sum.

Fibonacci Series:

- 1 Read the number of terms.
- 2 Initialize first = 0 and second = 1.
- 3 Display first.
- 4 Calculate the next term.
- 5 Update the values.
- 6 Repeat until the required number of terms is generated.

Part D – Methods**Factorial:**

- 1 Pass n to factorial(n).
- 2 Initialize fact = 1.
- 3 Multiply numbers from 1 to n.
- 4 Return fact.

Prime:

- 1 Pass n to isPrime(n).
- 2 Check divisibility from 2 to sqrt(n).
- 3 Return false if divisible.
- 4 Otherwise return true.

Maximum:

- 1 Pass two numbers to maximum(a, b).
- 2 Compare both numbers.
- 3 Return the larger number.

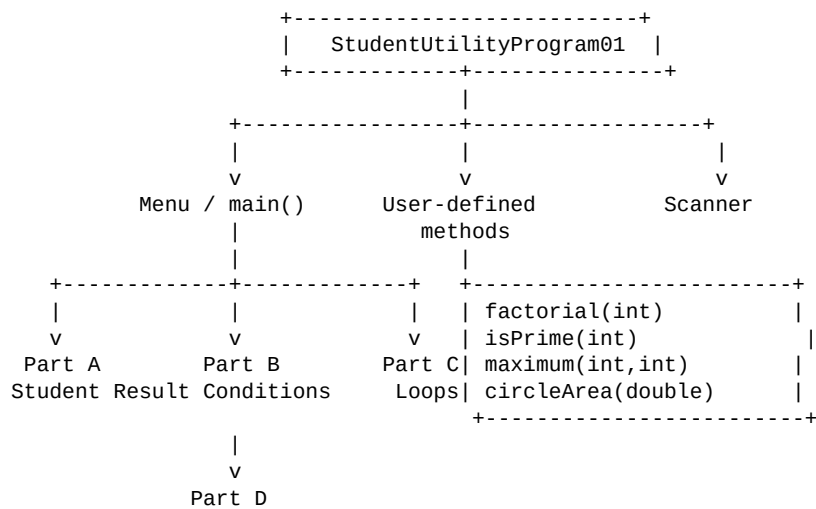
Circle Area:

- 1 Pass radius to circleArea(radius).
- 2 Calculate $\pi \times \text{radius} \times \text{radius}$.
- 3 Return the calculated area.

4. Exact UML Diagram for Your Code

```
+-----+
|               StudentUtilityProgram01               |
+-----+
| - sc : Scanner                                       |
+-----+
| + factorial(n : int) : long                          |
| + isPrime(n : int) : boolean                       |
| + maximum(a : int, b : int) : int                  |
| + circleArea(radius : double) : double             |
| + main(args : String[]) : void                     |
+-----+
```

Exact method usage in your code



5. Code

```
package packagecodejava;
import java.util.Scanner;

public class StudentUtilityProgram01 {

    // Scanner object for input
    static Scanner sc = new Scanner(System.in);

    // =====
    // PART D: USER-DEFINED METHODS
    // =====

    // Method to find factorial
    public static long factorial(int n) {
        long fact = 1;

        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }

        return fact;
    }

    // Method to check whether a number is prime
    public static boolean isPrime(int n) {
        if (n <= 1) {
            return false;
        }

        for (int i = 2; i <= Math.sqrt(n); i++) {
            if (n % i == 0) {
                return false;
            }
        }

        return true;
    }

    // Method to find maximum of two numbers
    public static int maximum(int a, int b) {
        if (a > b) {
            return a;
        } else {
            return b;
        }
    }

    // Method to calculate area of circle
    public static double circleArea(double radius) {
        return Math.PI * radius * radius;
    }

    // =====
    // MAIN METHOD
    // =====

    public static void main(String[] args) {

        int choice;

        do {
            System.out.println("\n=====");
            System.out.println("          STUDENT UTILITY PROGRAM");
            System.out.println("=====");
            System.out.println("1. Student Information and Result");
            System.out.println("2. Check Even or Odd");
            System.out.println("3. Find Largest of Three Numbers");
            System.out.println("4. Display Grade");
            System.out.println("5. Display Day of the Week");
            System.out.println("6. Multiplication Table");
            System.out.println("7. Display Numbers from 1 to N");
            System.out.println("8. Sum of First N Natural Numbers");
```

```

System.out.println("9. Fibonacci Series");
System.out.println("10. Factorial");
System.out.println("11. Check Prime Number");
System.out.println("12. Maximum of Two Numbers");
System.out.println("13. Area of a Circle");
System.out.println("14. Exit");
System.out.println("=====");

System.out.print("Enter your choice: ");
choice = sc.nextInt();

switch (choice) {

    // =====
    // PART A: STUDENT INFORMATION
    // =====

    case 1:
        sc.nextLine(); // Clear input buffer

        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Roll Number: ");
        int rollNumber = sc.nextInt();

        System.out.print("Enter marks for Subject 1: ");
        double mark1 = sc.nextDouble();

        System.out.print("Enter marks for Subject 2: ");
        double mark2 = sc.nextDouble();

        System.out.print("Enter marks for Subject 3: ");
        double mark3 = sc.nextDouble();

        double total = mark1 + mark2 + mark3;
        double percentage = total / 3;

        System.out.println("\n----- STUDENT RESULT -----");
        System.out.println("Name      : " + name);
        System.out.println("Roll Number: " + rollNumber);
        System.out.println("Mark 1    : " + mark1);
        System.out.println("Mark 2    : " + mark2);
        System.out.println("Mark 3    : " + mark3);
        System.out.println("Total     : " + total);
        System.out.println("Percentage : " + percentage + "%");

        break;

    // =====
    // PART B: CONDITIONAL STATEMENTS
    // =====

    case 2:
        System.out.print("Enter a number: ");
        int evenOddNumber = sc.nextInt();

        if (evenOddNumber % 2 == 0) {
            System.out.println(evenOddNumber + " is Even.");
        } else {
            System.out.println(evenOddNumber + " is Odd.");
        }

        break;

    case 3:
        System.out.print("Enter first number: ");
        int num1 = sc.nextInt();

        System.out.print("Enter second number: ");
        int num2 = sc.nextInt();

        System.out.print("Enter third number: ");
        int num3 = sc.nextInt();
        int largest;

```

```

        if (num1 >= num2 && num1 >= num3) {
            largest = num1;
        } else if (num2 >= num1 && num2 >= num3) {
            largest = num2;
        } else {
            largest = num3;
        }

        System.out.println("Largest number = " + largest);

        break;

    case 4:
        System.out.print("Enter percentage: ");
        double percentageInput = sc.nextDouble();

        if (percentageInput >= 90) {
            System.out.println("Grade: A+");
        } else if (percentageInput >= 80) {
            System.out.println("Grade: A");
        } else if (percentageInput >= 70) {
            System.out.println("Grade: B");
        } else if (percentageInput >= 60) {
            System.out.println("Grade: C");
        } else if (percentageInput >= 50) {
            System.out.println("Grade: D");
        } else if (percentageInput >= 40) {
            System.out.println("Grade: E");
        } else {
            System.out.println("Grade: F - Fail");
        }

        break;

    case 5:
        System.out.print("Enter day number (1-7): ");
        int day = sc.nextInt();

        switch (day) {
            case 1:
                System.out.println("Monday");
                break;

            case 2:
                System.out.println("Tuesday");
                break;

            case 3:
                System.out.println("Wednesday");
                break;

            case 4:
                System.out.println("Thursday");
                break;

            case 5:
                System.out.println("Friday");
                break;

            case 6:
                System.out.println("Saturday");
                break;

            case 7:
                System.out.println("Sunday");
                break;

            default:
                System.out.println("Invalid day number.");
        }

        break;

    // =====
    // PART C: LOOPING STATEMENTS

```



```
// =====

case 6:
    System.out.print("Enter a number for multiplication table: ");
    int tableNumber = sc.nextInt();

    System.out.print("Enter the limit: ");
    int tableLimit = sc.nextInt();

    System.out.println("\nMultiplication Table of " + tableNumber);

    for (int i = 1; i <= tableLimit; i++) {
        System.out.println(
            tableNumber + " x " + i + " = " + (tableNumber * i)
        );
    }

    break;

case 7:
    System.out.print("Enter N: ");
    int n = sc.nextInt();

    if (n < 1) {
        System.out.println("Please enter a positive number.");
    } else {
        System.out.println("Numbers from 1 to " + n + ":");

        for (int i = 1; i <= n; i++) {
            System.out.print(i + " ");
        }

        System.out.println();
    }

    break;

case 8:
    System.out.print("Enter N: ");
    int sumN = sc.nextInt();

    if (sumN < 1) {
        System.out.println("Please enter a positive number.");
    } else {
        int sum = 0;
        int i = 1;

        while (i <= sumN) {
            sum = sum + i;
            i++;
        }

        System.out.println(
            "Sum of first " + sumN +
            " natural numbers = " + sum
        );
    }

    break;

case 9:
    System.out.print("Enter number of Fibonacci terms: ");
    int terms = sc.nextInt();

    if (terms <= 0) {
        System.out.println("Please enter a positive number.");
    } else {
        int first = 0;
        int second = 1;

        System.out.println("Fibonacci Series:");

        int count = 1;

        do {

```

```

        System.out.print(first + " ");

        int next = first + second;
        first = second;
        second = next;

        count++;

    } while (count <= terms);

    System.out.println();
}

break;

// =====
// PART D: METHODS
// =====

case 10:
    System.out.print("Enter a number: ");
    int factorialNumber = sc.nextInt();

    if (factorialNumber < 0) {
        System.out.println("Factorial is not defined for negative numbers.");
    } else if (factorialNumber > 20) {
        System.out.println(
            "Please enter a number from 0 to 20."
        );
    } else {
        long result = factorial(factorialNumber);

        System.out.println(
            "Factorial of " + factorialNumber +
            " = " + result
        );
    }

    break;

case 11:
    System.out.print("Enter a number: ");
    int primeNumber = sc.nextInt();

    if (isPrime(primeNumber)) {
        System.out.println(primeNumber + " is a Prime number.");
    } else {
        System.out.println(primeNumber + " is not a Prime number.");
    }

    break;

case 12:
    System.out.print("Enter first number: ");
    int firstNumber = sc.nextInt();

    System.out.print("Enter second number: ");
    int secondNumber = sc.nextInt();

    int maximumNumber = maximum(firstNumber, secondNumber);

    System.out.println(
        "Maximum number = " + maximumNumber
    );

    break;

case 13:
    System.out.print("Enter radius of the circle: ");
    double radius = sc.nextDouble();

    if (radius < 0) {
        System.out.println("Radius cannot be negative.");
    } else {
        double area = circleArea(radius);

```

```
        System.out.println(
            "Area of circle = " + area
        );
    }

    break;

// =====
// EXIT
// =====

case 14:
    System.out.println("\nThank you for using Student Utility Program!");
    break;

default:
    System.out.println("Invalid choice. Please enter 1 to 14.");
}

} while (choice != 14);

sc.close();
}
}
```

6. Output

Complete sample execution

```
=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit
=====
Enter your choice: 1
Enter Student Name: seshu
Enter Roll Number: 218
Enter marks for Subject 1: 90
Enter marks for Subject 2: 80
Enter marks for Subject 3: 90

----- STUDENT RESULT -----
Name      : seshu
Roll Number: 218
Mark 1    : 90.0
Mark 2    : 80.0
Mark 3    : 90.0
Total     : 260.0
Percentage : 86.6666666666667%

=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit
=====
Enter your choice: 2
Enter a number: 10
10 is Even.

=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit
=====
Enter your choice: 3
Enter first number: 10
Enter second number: 25
Enter third number: 15
Largest number = 25

=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
```

2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit

=====
Enter your choice: 4
Enter percentage: 86
Grade: A

=====
STUDENT UTILITY PROGRAM
=====

1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit

=====
Enter your choice: 5
Enter day number (1-7): 3
Wednesday

=====
STUDENT UTILITY PROGRAM
=====

1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit

=====
Enter your choice: 6
Enter a number for multiplication table: 5
Enter the limit: 5

Multiplication Table of 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25

=====
STUDENT UTILITY PROGRAM
=====

1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit

=====
Enter your choice: 7
Enter N: 5

Numbers from 1 to 5:

1 2 3 4 5

```
=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit
=====
Enter your choice: 8
Enter N: 5
Sum of first 5 natural numbers = 15
```

```
=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit
=====
Enter your choice: 9
Enter number of Fibonacci terms: 7
Fibonacci Series:
0 1 1 2 3 5 8
```

```
=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit
=====
Enter your choice: 10
Enter a number: 5
Factorial of 5 = 120
```

```
=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit
=====
Enter your choice: 11
```

Enter a number: 17
17 is a Prime number.

```
=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit
=====
Enter your choice: 12
Enter first number: 25
Enter second number: 40
Maximum number = 40
```

```
=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit
=====
Enter your choice: 13
Enter radius of the circle: 5
Area of circle = 78.53981633974483
```

```
=====
      STUDENT UTILITY PROGRAM
=====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime Number
12. Maximum of Two Numbers
13. Area of a Circle
14. Exit
=====
Enter your choice: 14
```

Thank you for using Student Utility Program!